ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** the covered half remains brown with iodine	Light needed.
2	**(c)** light energy to chemical energy	Stored in glucose.
3	**(b)** nitrogen-fixing bacteria	Rhizobium.
4	**(b)** chlorophyll is needed	Pigment.
5	**(b)** chemical energy to usable energy	Released.
6	**(c)** nitrogen left by pulses	Rhizobium.
7	**(d)** NaOH absorbs carbon dioxide	No CO2.
8	**(d)** more glucose than used in respiration	Growth.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The green pigment traps light energy, which explains why chlorophyll is needed.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but guard cells do not explain why gases exchange occurs through stomata.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The plant uses CO2 produced and gives oxygen (1); The candle can burn longer (1).	1+1
12	Lack of oxygen (1); They cannot respire (1).	1+1
13	Spines and thick cuticle (1); Fewer stomata (1).	1+1
14	No light for photosynthesis (1); Stored starch was used (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hydrilla in water under funnel (1); Test tube collects gas (1); Control in dark (1).	1+1+1
16	Light is available at the top (1); Competition for sunlight (1); More photosynthesis (1).	1+1+1
17	Faster photosynthesis (1); More oxygen (1); Gas escapes (1).	1+1+1
	OR: Starch increases through the day (1); Photosynthesis exceeds respiration (1); Falls at night (1).	1+1+1
18	They absorb carbon dioxide (1); They release oxygen (1); They trap dust on their leaves (1).	1+1+1
	OR: Large in shade (1); Small or spines in desert (1); Floating in ponds (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) To use up (remove) the stored starch from the leaf (1). (ii) Iodine solution (1). (iii) Blue-black (1). (iv) The covered part does not turn blue-black (1), which shows that light is necessary for photosynthesis (starch formation) (1).	1+1+1+2
20	Parasite or saprotroph (1); Check connection to living tissue (1); Observe on dead wood (1); Test for starch (1); Conclude (1).	1+1+1+1+1
	OR: Photosynthesis takes CO2, gives O2 (1); Respiration reverse (1); Balance in nature (1); Burning adds CO2 (1); Plants reduce it (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that photosynthesis takes place only at night, or that respiration takes place only at night; photosynthesis needs **light**, respiration goes on **all the time**.
- Confusing xylem and phloem: **xylem** = water and minerals up, **phloem** = food from leaves.
- Writing that oxygen is a raw material of photosynthesis; the raw materials are carbon dioxide and water, and oxygen is the *by-product*.
- Mixing up transpiration (loss of water vapour through stomata) with absorption of water by the root hairs.
- Writing that Cuscuta makes its own food; it is a **parasite** without chlorophyll.
- Forgetting that iodine test shows **starch** (blue-black), not glucose, and that the leaf should first be boiled in alcohol to remove chlorophyll.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** it gains mass and grows	Balanced gas exchange.
2	**(c)** saprotroph	Feeds on bread.
3	**(d)** high humidity	Humid air reduces.
4	**(a)** photosynthesis is less than respiration	Low light.
5	**(d)** Cuscuta, a parasite	Lack chlorophyll.
6	**(a)** a plant with many large leaves	More surface for water loss.
7	**(c)** nitrogen or magnesium	Chlorophyll needs them.
8	**(a)** insectivorous	Needs nitrogen.
9	**(c)** A is true but R is false.	A is true; transpiration occurs mainly through stomata in leaves, so R is false.
10	**(d)** A is false but R is true.	Xylem carries water and minerals, so A is false; phloem carries food, so R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Dissolves chlorophyll (1); Makes colour change clearer (1).	1+1
12	Alga provides food (1); Fungus provides water and protection (1).	1+1
13	Soil is poor in nitrogen (1); They trap insects for nitrogen (1).	1+1
14	Photosynthesis is faster than respiration (1); Net release of oxygen (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Raise photosynthesis (1); Faster growth (1); Higher yields (1).	1+1+1
16	Fixes nitrogen to nitrates (1); Plant makes amino acids (1); Proteins formed (1).	1+1+1
17	First in food chain (1); Make food from sunlight (1); Support consumers (1).	1+1+1
	OR: Magnesium needed for chlorophyll (1); Less chlorophyll (1); Pale leaves (1).	1+1+1
18	They respond to light (1); More light for food making (1); Plants respond to stimuli (1).	1+1+1
	OR: Place a stem in coloured water (1); Colour rises through xylem (1); Cut section shows coloured vessels (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Transpiration (1). (ii) Stomata (1). (iii) Xylem (1). (iv) It cools the plant (1) and helps pull water and minerals upward from the roots (1).	1+1+1+2
20	Less photosynthesis (1); Less transpiration so less rain (1); More CO2 (1); Tree planting (1); Green belts (1).	1+1+1+1+1
	OR: Modified leaves (1); Digestive juices (1); They still photosynthesise (1); Need extra nitrogen (1); Rooted, have chlorophyll (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that photosynthesis takes place only at night, or that respiration takes place only at night; photosynthesis needs **light**, respiration goes on **all the time**.
- Confusing xylem and phloem: **xylem** = water and minerals up, **phloem** = food from leaves.
- Writing that oxygen is a raw material of photosynthesis; the raw materials are carbon dioxide and water, and oxygen is the *by-product*.
- Mixing up transpiration (loss of water vapour through stomata) with absorption of water by the root hairs.
- Writing that Cuscuta makes its own food; it is a **parasite** without chlorophyll.
- Forgetting that iodine test shows **starch** (blue-black), not glucose, and that the leaf should first be boiled in alcohol to remove chlorophyll.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Life Processes in Plants** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** magnesium	Magnesium is part of chlorophyll.
2	**(a)** oxygen	Gas.
3	**(b)** thick cuticle	Waxy cuticle.
4	**(a)** green plants	Producers.
5	**(a)** alcohol	They are sensitive to pollutants.
6	**(a)** the green stem	Leaves are spines.
7	**(a)** the Sun	Light energy.
8	**(c)** roots absorb dissolved minerals and need oxygen	Respiration needed.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Rhizobium fixes nitrogen for the plant, so less fertiliser is needed.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true statements about non-green or special nutrition, but they are unrelated.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Adds nitrogen (1); Improves soil (1).	1+1
12	They grow towards light (1); More light reaches the leaves for making food (1).	1+1
13	Dust blocks light and stomata (1); Less photosynthesis (1).	1+1
14	Reduce water loss (1); Protected from direct sunlight (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Leaves catch light and gases (1); Root hairs absorb water (1); Large area helps both (1).	1+1+1
16	Close stomata (1); Prevent water loss (1); No photosynthesis at night (1).	1+1+1
17	Starch in the cotyledons (1); Used for growth (1); Until leaves make food (1).	1+1+1
	OR: After dark: no starch (1); After light: starch present (1); Photosynthesis restarts (1).	1+1+1
18	Respiration cannot occur (1); No photosynthesis (1); Plant dies (1).	1+1+1
	OR: Legume with Rhizobium (1); Lichen with fungus (1); Parasites on hosts (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Parasitic nutrition (1). (ii) Cuscuta (1). (iii) Rhizobium (1). (iv) The pitcher plant grows in nitrogen-poor soil and obtains nitrogen from insects (1); Cuscuta takes ready-made food from the hedge plant, leaving less food for it (1).	1+1+1+2
20	Adequate sunlight (1); Crop rotation with legumes (1); Irrigation scheduling (1); Spacing of plants (1); Mulching (1).	1+1+1+1+1
	OR: Sunny spot (1); Compost (1); Pulses in rotation (1); Mulch to reduce transpiration (1); Regular watering (1).	1+1+1+1+1

Mistakes to avoid:

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- Confusing xylem and phloem: **xylem** = water and minerals up, **phloem** = food from leaves.
- Writing that oxygen is a raw material of photosynthesis; the raw materials are carbon dioxide and water, and oxygen is the *by-product*.
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- Writing that Cuscuta makes its own food; it is a **parasite** without chlorophyll.
- Forgetting that iodine test shows **starch** (blue-black), not glucose, and that the leaf should first be boiled in alcohol to remove chlorophyll.

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